white-noise term with the following stochastic property

$$\langle \theta_{ij}(t) \rangle = 0, \qquad \langle \theta_{ij}(t)\theta_{kl}(t') \rangle = (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})\delta(t - t'), \tag{8.88}$$

and are independent for each pair of interacting particles at each time step. Invariance of the canonical distribution under these equations of motion follows if

$$\omega^{\mathrm{D}}(r_{ij}) = [\omega^{\mathrm{R}}(r_{ij})]^2, \quad \sigma^2 = 2\gamma \mathrm{k_B T}, \tag{8.89}$$

where $\mathrm{k_B}$ is the Boltzmann constant and T the equilibrium temperature. Note that the "canonical distribution" in the case of the DPD system or other momentum-conserving thermostat should have a density which is reduced to the leaves of constant total momentum, i.e.

$$\hat{\rho}_\beta \stackrel{\mathrm{def}}{=} Z^{-1} e^{-\beta H(\boldsymbol{q},\boldsymbol{p})} \times \delta \left[\sum_{i=1}^{\mathrm{N}} p_{i,x} - \pi_x \right] \delta \left[\sum_{i=1}^{\mathrm{N}} p_{i,y} - \pi_y \right] \delta \left[\sum_{i=1}^{\mathrm{N}} p_{i,z} - \pi_z \right]$$

where $\boldsymbol{\pi} = (\pi_x, \pi_y, \pi_z)$ is the total momentum vector.

The conservation of the total momentum is a direct result of the symmetric form of the various forces, thus DPD may be viewed as an isotropic Galilean-invariant thermostat which preserves hydrodynamics [8]. The ergodicity of DPD (i.e. that $\hat{\rho}_\beta$ is the unique measure sampled by the dynamics) has only been demonstrated in the case of a high density of particles in one dimension [334].

The DPD system may be viewed as a special case of the generalized Langevin dynamics with equations

$$\begin{aligned} \mathrm{d}\boldsymbol{q} &= \boldsymbol{M}^{-1}\boldsymbol{p}\mathrm{d}t, \\ \mathrm{d}\boldsymbol{p} &= -\nabla U(\boldsymbol{q})\mathrm{d}t - \boldsymbol{\Delta}(\boldsymbol{q})\boldsymbol{p}\mathrm{d}t + \boldsymbol{\Sigma}(\boldsymbol{q})\mathrm{d}\mathbf{W}, \end{aligned} \tag{8.90}$$

where the potential U is given in (8.84), and the damping matrix $\boldsymbol{\Delta}$ and stochastic multiplier matrix $\boldsymbol{\Sigma}$ are related by

$$\boldsymbol{\Delta} = \frac{\boldsymbol{M}^{-1}}{2\mathrm{k_B T}} \boldsymbol{\Sigma}\boldsymbol{\Sigma}^T.$$

In the next subsection we describe the numerical treatment of the DPD system. We then take up the alternative scheme, the pairwise Nosé-Hoover-Langevin (PNHL) method, of [228] which also conserves momentum exactly. Other alternatives to the DPD system include the Peters thermostat [299], the Lowe-Andersen method [241], and the Nosé-Hoover-Lowe-Andersen thermostat [347]. We do not discuss these schemes here, but note that they are compared to DPD and some of the other schemes introduced below in [228].

8.9.2 Numerical Treatment of the DPD System

In comparing the DPD system to standard molecular dynamics, a key feature is the "soft" nature of the repulsive potential U_{DPD} defined by (8.84). The lack of any sort of stiff harmonic component or steep repulsive potentials such as those present in typical molecular models means that the stepsize is not dominated by the stability restriction, but rather by accuracy requirements. That is, DPD simulations may be stable for stepsizes which lead to very large errors in thermodynamic averages.

Many efforts have been made to develop accurate integration schemes for the DPD system. With few exceptions, the methods used are variants of the velocity-Verlet integrator, owing to its long history in MD. Some early integration schemes are to be found in the papers of Groot and Warren [157], Gibson et al. [148], and Besold et al. [34, 385]. The latter work describes the so-called DPD-Velocity Verlet (DPD-VV) method. Unfortunately [34, 385], all of these methods exhibit significant simulation errors for large stepsize. Much of the error can be seen as the result of requiring an explicit treatment of the dissipative terms, for reasons of computational efficiency. More accurate schemes are possible by using implicit (or semi-implicit) methods such the self-consistent leap-frog integrator (see [34, 295]). However, these methods are typically much slower than the simpler, explicit methods [73, 281], the consequence being that the simpler DPD-VV is most often used (with a relatively small timestep). Further alternatives to the DPD-VV method include a scheme of den Otter and Clarke [101], splitting methods (Shardlow [333], De Fabritiis et al. [96] and Thalmann and Farago [363]), and multiple time step schemes of Jakobsen et al. [191].

8.9.2.1 DPD Velocity-Verlet

With integration stepsize h, the DPD-VV method may be written as

$$\boldsymbol{p}_{n+1/2} = \boldsymbol{p}_n + \frac{h}{2}\boldsymbol{F}^C(\boldsymbol{q}_n) + \frac{h}{2}\boldsymbol{F}^D(\boldsymbol{q}_n, \boldsymbol{p}_n) + \frac{\sqrt{h}}{2}\boldsymbol{F}^R(\boldsymbol{q}_n), \tag{8.91}$$

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2}, \tag{8.92}$$

$$\boldsymbol{p}_{n+1} = \boldsymbol{p}_{n+1/2} + \frac{h}{2}\boldsymbol{F}^C(\boldsymbol{q}_{n+1}) + \frac{h}{2}\boldsymbol{F}^D(\boldsymbol{q}_{n+1}, \boldsymbol{p}_{n+1/2}) + \frac{\sqrt{h}}{2}\boldsymbol{F}^R(\boldsymbol{q}_{n+1}). \tag{8.93}$$

In this scheme, the conservative forces are recalculated once per timestep (the conservative force computed may be re-used at the following timestep). The dissipative force must be computed twice, and the new random force used in (8.93) should also be used in the subsequent step. This scheme is not symmetric for

deterministic systems and would be expected to be only first order accurate. It could be made symmetric by a simple change, for example replacing $\boldsymbol{p}_n$ in the force computation at the first step by $\boldsymbol{p}_{n+1/2}$, but these changes would make the method implicit.

8.9.2.2 Shardlow's Splitting Method

Shardlow [333] proposed splitting methods for DPD based on dividing the force field of the DPD system into its separate (conservative+dissipative+random) parts. Several integrators are then derived based on approximating the various terms. In the most popular and efficient variant, the scheme of Brünger, Brooks and Karplus (BBK) [55] is used to solve the dissipative+stochastic terms, with standard velocity Verlet applied to the conservative part.

We first separate the system of stochastic differential equations for DPD (8.90) into three pieces, which we label as A, B and O:

$$\mathrm{d}\begin{bmatrix}\boldsymbol{q}_i\\ \boldsymbol{p}_i\end{bmatrix} = \underbrace{\begin{bmatrix} m_i^{-1}\boldsymbol{p}_i\\ \boldsymbol{0}\end{bmatrix}\mathrm{d}t}_{\mathrm{A}} + \underbrace{\begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{F}_i^{\mathrm{C}}\end{bmatrix}\mathrm{d}t}_{\mathrm{B}} + \underbrace{\begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{F}_i^{\mathrm{D}}\mathrm{d}t + \sigma\boldsymbol{R}_i\end{bmatrix}}_{\mathrm{O}}, \tag{8.94}$$

where $\boldsymbol{R}_i = \sum_{j\neq i}\omega^{\mathrm{R}}(r_{ij})\hat{\boldsymbol{q}}_{ij}\mathrm{dW}_{ij}$, for standard i.i.d. Wiener processes $\mathrm{W}_{ij}(t)$, such that $\mathrm{dW}_{ij} = \mathrm{dW}_{ji}$. The generators for each part of the SDE may be written out as follows:

$$\mathcal{L}_{\mathrm{A}} = \sum_i \frac{\boldsymbol{p}_i}{m_i}\cdot\nabla_{\boldsymbol{q}_i}, \tag{8.95}$$

$$\mathcal{L}_{\mathrm{B}} = \sum_i \boldsymbol{F}_i^{\mathrm{C}}\cdot\nabla_{\boldsymbol{p}_i} = -\sum_i \nabla_{\boldsymbol{q}_i}U(\boldsymbol{q})\cdot\nabla_{\boldsymbol{p}_i}, \tag{8.96}$$

$$\mathcal{L}_{\mathrm{O}} = \sum_i\sum_{j\neq i}\left(-\gamma\omega^{\mathrm{D}}(r_{ij})(\hat{\boldsymbol{q}}_{ij}\cdot\boldsymbol{v}_{ij}) + \frac{\sigma^2}{2}[\omega^{\mathrm{R}}(r_{ij})]^2\hat{\boldsymbol{q}}_{ij}\cdot\left[\nabla_{\boldsymbol{p}_i}-\nabla_{\boldsymbol{p}_j}\right]\right)\hat{\boldsymbol{q}}_{ij}\cdot\nabla_{\boldsymbol{p}_i}. \tag{8.97}$$

Thus, the generator for the DPD system can be written as

$$\mathcal{L}_{\mathrm{DPD}} = \mathcal{L}_{\mathrm{A}} + \mathcal{L}_{\mathrm{B}} + \mathcal{L}_{\mathrm{O}}, \tag{8.98}$$

and the flow map may be written in the standard shorthand as

$$\mathscr{F}_t = e^{t\mathcal{L}_{\mathrm{DPD}}}.$$

Approximations of $\mathscr{F}_t$ can be constructed as for Langevin dynamics.